

COLLEGE CODERS

React JS Lesson 13 – Props Drilling (Premium Notes)

1) What Props Drilling Actually Means

- Props drilling means passing data through multiple components even when middle components do not use it.
- Data travels parent → child → child → child.

Example:

```
<Parent data={name} />
<Child1 data={name} />
<Child2 data={name} />
```

Explanation:

- Each level forwards same prop.

2) Why it becomes a problem in large apps

- Too many intermediate components make code harder to manage.
- Changes become difficult in deep trees.

Example:

Parent → Child1 → Child2 → Child3 → Child4

Explanation:

- Every layer must pass same prop.

3) Simple practical React code explanation

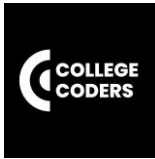
- Parent sends prop, middle child forwards it.

Example:

```
function Parent(){
  return <Child1 name="Rohith" />
}
```

```
function Child1({name}){
  return <Child2 name={name} />
}
```

```
function Child2({name}){
```



COLLEGE CODERS

```
return <h2>{name}</h2>
}
```

Explanation:

- Child1 only forwards prop without using it.

4) When to use props drilling and when to avoid it

- Use props drilling for small component depth.
- Avoid when too many nested layers exist.

Example:

2 levels → okay
5+ levels → avoid

Explanation:

- Context API becomes better in deep trees.

5) Best practices for clean component structure

- Keep component tree shallow.
- Pass only required props.
- Use Context API when shared state grows.

Example:

Clean structure:
Parent → Child → Final Child

Explanation:

- Avoid unnecessary middle forwarding.

Mini Project – Passing User Name Through Components

- Parent stores user name.
- Pass through two child levels.
- Display in final child.